Bootcamp Data Discoveries

What Senzing found in the data the bootcamp generated

Customer 360 scenario: CRM_SYSTEM, BILLING_SYSTEM and SUPPORT_TICKETS, resolved with the Senzing SDK. Every figure below comes from the SDK (export stream, get_entity, why_records, how_entity, find_network, find_path) run by the programs in src/query/.

Your question, answered

Requirement	Answer	Where in this document
Which customers appear in all three systems, and what does their unified record look like?	451 customers appear in all three systems; each unified record merges one name, address, phone and email from the three systems (e.g. Kyle Yorkbrook, entity 300009: BIL-000100, CRM-000100, CRM-003782 and TKT-000100).	Merges and match keys
Which records are likely duplicates of each other but weren't merged, so they need a human review?	737 possibly-same pairs (1,336 entities), 679 of them across systems; every pair matched on name only (+NAME). Against the answer key 644 are the same person and 93 are different people.	Review queue
How many unique customers do we really have, compared with the raw record count in each system?	6,671 unique customers behind 10,000 records (1.5 records per customer): CRM 4,000 records / 3,702 customers, Billing 4,000 / 3,923, Support 2,000 / 1,688.	Headline numbers, interpreted

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- 10,000 records loaded, 6,671 entities resolved. One customer in three has records in more than one place, so a third of the "customers" in any one system report is a duplicate view of someone else's.
- 2,460 entities hold more than one record; 2,191 span two or more systems; 451 span all three. These are the customers the business could not see whole before.
- **Within-system duplicates:** CRM 298 (campaign re-entries), Support 312 (repeat tickets), Billing 77.
- **Accuracy against the scenario's answer key:** precision 100% (no wrong merges), recall 80.9%. Every merge Senzing made was right; about one true match in five was left as a possible match rather than merged.

Merges and match keys

The match keys behind the merges (record joins, by rule):

Match key	Rule	Records joined
+NAME+ADDRESS+PHONE	MFF_CNAME	1,064
+NAME+EMAIL	SF1_CNAME	921
+NAME+ADDRESS	CNAME_CFF	764
+NAME+PHONE	CNAME_CFF	263
+EXACTLY_SAME	(exact duplicate)	157
+NAME+ADDRESS+PHONE+EMAIL	SF1_PNAME_CFF	79

Billing joins CRM mostly on name + address + phone; Support joins CRM on name + email. The full list of 451 three-system customers is in data/query_results/customers_in_all_sources.json.

Headline entity: Alan Whitevale (entity 400155), 5 records from all three systems: two CRM records sharing name, address, phone and email; two Support tickets joined on +NAME+EMAIL; one Billing

record joined on +NAME+ADDRESS despite "Ln" vs "Lane".

Review queue

737 possibly-same pairs, all on +NAME (full queue: data/query_results/review_queue.json). By system pair: Support vs Billing 204, CRM vs Billing 165, CRM+Billing vs Support 106, CRM vs Support 67, Support vs Support 41.

- **Robert Rainridge, entities 300005 and 801540:** CRM has street, phone and email; Billing BIL-001046 has only name and "Duluth, MN 55805". Same person, nothing to confirm it beyond the name.
- **Mary Huntcroft (300039) and "Mary" (801469):** Billing BIL-002893 carries only the first name "Mary" in Des Moines; Mary Huntcroft lives in Albany. Different people, correctly not merged.

Each row is one human decision away from being acted on; a shared customer number across systems would settle most of them automatically.

Why and how: worked examples

Why (why_records), Yvonne Vanceson, entity 300010:

compared directly, Support TKT-000383 (name, email) and Billing BIL-000383 (name, address, phone) share only the name: why key +NAME, rule SNAME, POSSIBLY_SAME, name score 100.

Near-miss (the one that teaches more):

on their own, those two records would never merge, because a matching name is not enough evidence. They sit in one entity only because CRM-000383 carries all four features and bridges them.

How (how_entity), same entity: Step 1, CRM-000383 + BIL-000383 on +NAME+ADDRESS+PHONE (MFF_CNAME; address 99 for "Blvd" vs "Boulevard", phone 100). Step 2, TKT-000383 joins that pair on +NAME+EMAIL (SF1_CNAME; email 100). Final state: one settled entity, nothing to re-evaluate.

Relationship networks

All relationships are discovered (possibly same 737, possibly related 10); none are disclosed in the source data.

- **The "Brad" cluster:** find_network from 400514 (Brad Crestson) and 401866 (Brad Highley) returned 8 entities and 13 possibly-same +NAME links. Both entities hold a Billing record with only NAME_FIRST "Brad" (BIL-000342, BIL-001739), so the bare first name links them to every other Brad.
- **A 2-degree path:** 400623 Harry Glenwell -> 400132 Harold Fairberry -> 400627 Harry Nashton, +NAME at each hop. Harold's Billing record is first-name-only, and "Harry" is a nickname of Harold, so the path is built on first names alone.

Finding: first-name-only Billing records create noisy links; they are a data-quality item for BILLING_SYSTEM, not real connections.

What was not found, and why

Measurement:

> 646 of 6,000 invented people remain split across entities (recall 80.9%). Their records share only a name: BILLING_SYSTEM has no email field and, after regeneration, about 18% of its records lack an address and 36% a phone; SUPPORT_TICKETS carries only name and email. Data collection measured 40 invented people sharing a name with a different person (a lower bound).

Which case applies: the data had little shared evidence to find, not that the pipeline underperformed. The mapping carries every field the sources have (the cause check found the near-misses are not mapping-actionable), the match-key audit found no suppressed features (single-source 0 of 269 keys, cross-source 0 of 3,060, relationships 0 of 747), and the how-state audit found all 2,460 multi-record entities settled.